Online Appendix for "Why Retirees Don't Annuitize: A Unified Lifecycle Model"

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A Model Details

A.1 Health transition matrices

Health transitions are calibrated to the RAND HRS longitudinal file (waves 4–16), with the five self-reported health categories mapped to three states: Good (Excellent/Very Good), Fair (Good), and Poor (Fair/Poor). Transition probabilities are age-dependent, estimated in five-year age bands from 65–69 through 95+.

The transition matrices exhibit two standard features. First, health deterioration is more likely than improvement: transition rates from Good to Fair exceed those from Fair to Good at all ages. Second, deterioration accelerates with age: the probability of remaining in Good health falls from approximately 0.80 at ages 65–69 to 0.55 at ages 90+.

A.2 Medical expenditure calibration

Log out-of-pocket medical expenditures follow:

$$\ln m_t = \mu_m^{\text{base}} + g \cdot (t - 65) + \delta_\mu(H_t) + \left[\sigma_m^{\text{base}} + \delta_\sigma(H_t)\right] \cdot \varepsilon_t, \quad \varepsilon_t \sim N(0, 1),$$

where $\mu_m^{\text{base}} = 7.037$ is the log mean at age 65 for Fair health, $g = 0.0652$ is the annual growth rate in the log mean, $\delta_\mu(H)$ shifts the log mean by health state $[-0.5, 0, 0.7]$ for Good, Fair, and Poor respectively, $\sigma_m^{\text{base}} = 1.4$ is the base log standard deviation, and $\delta_\sigma(H)$ shifts the log standard deviation by health $[-0.2, 0, 0.2]$.

Table 1: Medical Expenditure Moments: Model vs. Target

Moment	Model	Target (Jones et al. 2018)
Mean OOP, age 70, Fair health	\$4,201	\$4,200
Mean OOP, age 100, Fair health	\$29,703	\$29,700
95th pctile, age 100, Fair health	\$111,502	\$111,200

The lognormal specification produces a right-skewed distribution consistent with the heavy tails observed in medical expenditure data. The log standard deviation of 1.4 implies a coefficient of variation of approximately 2.5, which matches the empirical distribution of out-of-pocket spending in the HRS and MEPS.

A.3 Survival calibration

Baseline survival probabilities use the SSA administrative period life table employed by Lockwood (2012). The cumulative death probabilities at ages 65, 70, 75, ... are taken directly from Lockwood's replication code (BAP_sim2.m) and interpolated linearly at intermediate ages.

Health-adjusted survival follows the proportional hazard specification $s(t, H) = s_{\text{base}}(t)^{\mu(H)}$, where $\mu(H)$ is the health-specific hazard multiplier. At baseline, $\mu = [0.50, 1.0, 3.0]$. This implies that an individual in Good health at age 80—where $s_{\text{base}}(80) \approx 0.96$—survives with probability $0.96^{0.5} \approx 0.98$, while an individual in Poor health survives with probability $0.96^{3.0} \approx 0.89$.

The constant-multiplier specification is a simplification. The HRS data show that the health-mortality gradient compresses with age: the ratio of Poor-to-Fair mortality is 3.29 at ages 65–74, 2.77 at 75–84, and 1.82 at 85+. Age-varying multipliers would slightly atten-

uate the Reichling–Smetters mechanism at advanced ages. We use constant multipliers for tractability and because the annuitization decision is made at age 65, when the gradient is widest.

B Grid Convergence

Table 2 reports predicted ownership under the full model for different grid resolutions. The three grids—wealth (n_W), annuity income (n_A), and annuitization fraction (n_α)—have partially offsetting effects on ownership. Coarser annuity income grids smooth over the dip in value at intermediate α, biasing ownership upward. Coarser α grids miss beneficial annuitization fractions, biasing ownership downward. Both converge as resolution increases.

Table 2: Grid Convergence

Resolution	$n_W \times n_A \times n_\alpha$	Ownership (%)	Δ from production
Coarse	$30 \times 10 \times 21$	22.1	+20.7 pp
	$40 \times 15 \times 31$	13.6	+12.2 pp
Medium	$60 \times 20 \times 51$	6.5	+5.1 pp
	$60 \times 25 \times 101$	2.8	+1.4 pp
Production	$\mathbf{80 \times 30 \times 101}$	**1.4**	—
Fine	$80 \times 30 \times 201$	1.6	+0.2 pp
	$100 \times 40 \times 101$	1.4	0.0 pp
	$100 \times 40 \times 201$	1.4	0.0 pp

All specifications use baseline parameters ($\gamma = 2.5$, DFJ bequests, MWR $= 0.82$, $\pi = 2\%$, hazard multipliers $[0.50, 1.0, 3.0]$).

The production grid ($80 \times 30 \times 101$) yields 1.4% ownership. Doubling the wealth and annuity grids to 100×40 while holding $n_\alpha = 101$ produces the same 1.4%. Doubling n_α to 201 also produces 1.4%. Convergence within ± 0.2 percentage points is achieved at the production resolution.

C Bequest Specification

The De Nardi–French–Jones (DFJ) bequest utility is $v(b) = \theta(b+\kappa)^{1-\gamma}/(1-\gamma)$. The parameter κ controls the luxury-good nature of bequests. When κ is large relative to wealth b, marginal bequest utility $v'(b) = \theta(b+\kappa)^{-\gamma}$ is nearly constant across wealth levels, making bequests a luxury good: the bequest-to-wealth ratio rises with wealth.

The baseline parameters ($\theta = 56.96$, $\kappa = \$272{,}628$) are from Lockwood (2012), who estimated them from HRS data under the DFJ specification. At these values, an individual with \$50,000 in wealth has $v'(b) \approx \theta \cdot (322{,}628)^{-2.5}$, which is negligible compared to the marginal utility of consumption. The bequest motive binds meaningfully only for individuals with wealth approaching κ.

Homothetic bequest anomaly. Combining the DFJ bequest intensity $\theta = 56.96$ with a homothetic specification ($\kappa = \$10$) produces anomalous results. At $\kappa = \$10$, marginal bequest utility near $b = 0$ is $\theta \cdot 10^{-2.5} = 56.96 \times 3.16 \times 10^{-3} \approx 0.18$, which is extreme relative to marginal consumption utility. This creates a narrow precautionary buffer—individuals avoid annuitizing not to protect bequests, but to avoid the near-zero-wealth states where $v'(b)$ diverges. The predicted ownership under this misspecification (60.8%) is higher than the no-bequest case (15.7%), because the buffer effect dominates the bequest effect.

This anomaly illustrates why θ and κ must be interpreted as a pair, not independently. The DFJ estimates from Lockwood (2012) are jointly calibrated; using one with the other's value replaced is not a valid counterfactual.

D Comparison with Pashchenko (2013)

Pashchenko (2013) conducted a sequential decomposition with four channels: pre-existing annuitization (Social Security and DB pensions), bequest motives, minimum purchase requirements, and pricing loads. Her full model predicted approximately 20% participation,

roughly four times the observed rate of 5%.

We replicate her channel set within our framework. Our replication differs from hers in three respects: we use DFJ luxury-good bequests rather than the homothetic specification she employed; we use a continuous annuitization fraction rather than a binary purchase decision; and we do not model housing illiquidity. Despite these differences, the qualitative pattern matches.

Table 3: Pashchenko (2013) Replication and Extension

Model Specification	Ownership (%)	Δ
Yaari benchmark (SS on)	86.2	—
+ Bequest motives (DFJ)	86.0	-0.2 pp
+ Minimum purchase ($25K$)	70.7	-15.3 pp
+ Pricing loads (MWR=0.85)	66.6	-4.1 pp
Channels omitted by Pashchenko:		
+ Medical costs + R-S correlation	48.0	-18.6 pp
+ Inflation erosion (2%)	23.9	-24.1 pp
Observed (Lockwood 2012)	3.6	—

Steps 0–3 approximate Pashchenko's channels (SS, bequests, minimum purchase, pricing loads). Steps 4–5 add channels she omitted. Housing illiquidity is not modeled; see text for discussion.

Under Pashchenko's channels alone (Steps 0–3), our model predicts 66.6% ownership. The higher prediction relative to Pashchenko's 20% reflects differences in model specification, particularly the continuous annuitization margin. With a binary purchase decision and housing illiquidity (which Pashchenko includes and we do not), predicted participation would be lower.

The channels Pashchenko omitted—medical expenditure risk with health-mortality correlation (Step 4) and inflation erosion (Step 5)—reduce ownership from 66.6% to 23.9%. These two channels account for a larger ownership reduction (42.7 percentage points) than Pashchenko's four channels combined (19.5 percentage points from Steps 0 to 3).

E Deferred Income Annuity Extension

Deferred income annuities (DIAs) begin payments at a future age, typically 80 or 85. By concentrating payments in late life—when longevity risk is greatest—they offer higher per-dollar payouts. A DIA purchased at 65 with payments beginning at 80 pays roughly 2.8 times the per-period income of an immediate annuity (SPIA) per dollar of premium, and a DIA-85 pays roughly 5.9 times as much.

However, deferred products carry lower money's worth ratios. Wettstein et al. (2021) estimate MWRs of 0.50 for DIA-80 and 0.45 for DIA-85, compared to 0.82 for SPIAs. The lower MWR reflects the longer deferral period (more discounting), smaller market (less competition), and higher sensitivity to mortality projection errors.

Table 4: SPIA vs. Deferred Income Annuity (DIA) Comparison

	SPIA	DIA-80	DIA-85
MWR	0.82	0.50	0.45
Payout rate	0.0587	0.1659	0.3485
No bequest			
Ownership (%)	20.6	8.4	6.6
CEV at $200K$(%)	0.00	0.00	0.00
Optimal *alpha* at $200K$	0.0%	0.0%	0.0%
Moderate (DFJ)			
Ownership (%)	5.9	8.4	6.6
CEV at $200K$(%)	0.00	0.00	0.00
Optimal *alpha* at $200K$	0.0%	0.0%	0.0%

DIA MWR from Wettstein et al. (2021). All models include medical costs, R-S correlation, pricing loads, and 2% inflation. CEV evaluated at good health (H=1).

Table 4 compares predicted ownership and welfare across product types. All models include medical costs, health-mortality correlation, pricing loads, and 2% inflation.

Under no bequests, SPIA ownership is 20.6%, DIA-80 is 8.4%, and DIA-85 is 6.6%. The lower DIA ownership reflects the poor MWR: the actuarial advantage of deferred payout is more than offset by the pricing disadvantage. CEV at $200,000 wealth is zero for all three products.

With DFJ bequests, SPIA ownership falls to 5.9% while DIA ownership is unchanged at 8.4% (DIA-80) and 6.6% (DIA-85). DIAs are less affected by bequests because the premium is spent at age 65 but generates income only in late life, creating a natural separation between bequeathable wealth and annuity purchase.

The DIA results suggest that product innovation alone is unlikely to resolve low annuity demand. The channels suppressing SPIA demand—valuation risk, inflation erosion, bequest motives—apply with similar force to deferred products, and the inferior pricing of DIAs partially offsets their actuarial advantage.

F Full Sensitivity Analysis

Table 5 reports predicted ownership under all robustness specifications.

G Solution Algorithm

G.1 Backward induction

The model is solved by backward induction from the terminal period $T = 46$ (age 110) to $t = 1$ (age 65). At the terminal period, the individual consumes all available resources and leaves residual wealth as a bequest:

$$V(W, A, H, T) = \max_{c \leq W + A_T + \mathrm{SS}(T) - m_T} \Big\{ u(c) + v\Big((1+r)(W + A_T + \mathrm{SS}(T) - m_T - c)\Big) \Big\}.$$

For $t = T-1, \ldots, 1$, the value function is computed at each grid point (W_i, A_j, H_k) by solving the one-period consumption problem using Brent's method on the interval $[\underline{c},\ W + A_t + \mathrm{SS}(t) - m]$. The continuation value $V(W', A', H', t+1)$ is evaluated by piecewise linear interpolation over the wealth grid with flat extrapolation beyond the grid boundaries.

Table 5: Full Robustness Results

Category	Specification	Ownership (%)
Risk aversion (γ)		
	$\gamma = 1.50$	0.0
	$\gamma = 2.00$	0.0
	$\gamma = 2.30$	0.0
	$\gamma = 2.40$	0.0
	$\gamma = 2.45$	0.0
	$\gamma = 2.50$ (baseline)	1.4
	$\gamma = 2.55$	5.3
	$\gamma = 2.60$	6.9
	$\gamma = 2.70$	12.4
	$\gamma = 2.80$	17.1
	$\gamma = 3.00$	23.7
	$\gamma = 3.50$	31.6
	$\gamma = 4.00$	35.3
	$\gamma = 5.00$	39.6
Hazard multipliers		
	$[0.45, 1.0, 3.5]$ (R-S functional)	0.0
	$[0.50, 1.0, 3.0]$ (baseline)	1.4
	$[0.57, 1.0, 2.7]$ (HRS SRH)	5.7
	$[0.60, 1.0, 2.0]$ (conservative)	24.4
Minimum purchase threshold		
	\$0	1.4
	\$1,000	1.4
	\$5,000	1.4
	\$10,000	1.4
	\$25,000	1.4
Path dependence		
	Ordering A	1.4
	Ordering B	1.4
Grid convergence		
	Coarse (40×15)	13.6
	Medium (60×20)	6.5
	Production (80×30)	1.4
	Fine (100×40)	1.4
Bequest specification		
	No bequests ($\theta = 0$)	15.7
	DFJ luxury ($\theta = 56.96$, $\kappa =$ \$272K)	1.4

Unless otherwise noted, all specifications use baseline parameters: $\gamma = 2.5$, $\beta = 0.97$, DFJ bequests ($\theta = 56.96$, $\kappa =$ \$272,628), MWR $= 0.82$, $F =$ \$1,000, $\pi = 2\%$, hazard multipliers $[0.50, 1.0, 3.0]$, production grid ($80 \times 30 \times 101$).

G.2 Expectation computation

The expectation over next-period health and medical expenditures involves:

1. A sum over three health states weighted by transition probabilities $\pi(H_t, H', t)$.
2. A sum over five Gauss–Hermite quadrature nodes for the medical expenditure shock ε.

Each grid point requires $3 \times 5 = 15$ evaluations of the continuation value, yielding a total of $80 \times 30 \times 3 \times 46 \times 15 \approx 49.7$ million value function evaluations per model solve.

G.3 Annuitization optimization

At $t = 1$, the optimal annuitization fraction α^* is found by grid search over 101 equally spaced values in $[0, 1]$. For each candidate α, post-purchase wealth is $(1-\alpha)W_0$ and annuity income is $\alpha W_0 \times$ payout_rate. The value function at the resulting state is evaluated by bilinear interpolation over the (W, A) grid for the relevant health state. The fixed cost F is subtracted from value whenever $\alpha > 0$.

G.4 Population evaluation

Ownership is evaluated on a population of HRS individuals with observed initial wealth, permanent income, and age. For each individual, the optimal α^* is computed at their specific (W_0, y, H_0). Ownership is defined as $\alpha^* > 0$. The ownership rate is the fraction of the eligible population (wealth $\geq$ \$5,000) that annuitizes any positive amount.

G.5 Computational performance

A single model solve (one parameterization) requires approximately 60–120 seconds on an Apple M1 processor. The full decomposition (6 solves) runs in 6–12 minutes. The robustness

analysis (approximately 40 configurations) requires 1–2 hours. Grid convergence checks (8 configurations at varying resolutions) require 2–4 hours.

References

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